HUMAN FOLLOWING ROBOT

Under Supervision of: Priyadarshini R

SCHOOL: SENSE

TEAM MEMBERS:

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Introduction:

The human-following robot is a cutting-edge project aimed at creating an autonomous system that can detect and follow designated individuals . This robot leverages multiple sensing technologies, including IR sensors and an ultrasonic sensor, to navigate its environment effectively. By integrating these components with a WD car kit and a motor driver controlled by an Arduino, the robot can provide valuable assistance in various applications such as personal assistance, retail guidance, and caregiving.

Scope of the Project:

The scope of this project includes:

1.Technical Development:
1. Designing and assembling a robot capable of following a person .
2. Integrating 2 IR sensors and 1 ultrasonic sensor for comprehensive environment awareness.

2.Algorithm Implementation:
1. Developing algorithms for detecting and following a person while ensuring obstacle avoidance.
2. Utilizing basic control logic to enable smooth navigation.

3.Applications:
1. Exploring practical applications in healthcare, retail, and domestic environments to enhance user experience.

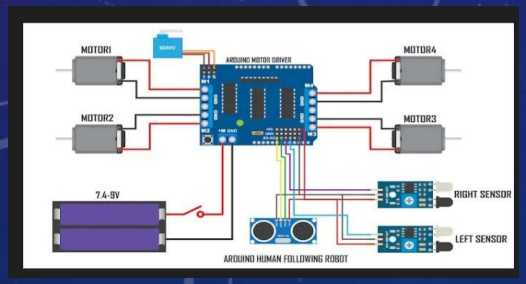
4.User Interaction:
1. Designing a simple user interface for easy operation, allowing users to start or stop the robot's following mode.
2. Collecting user feedback to enhance functionality

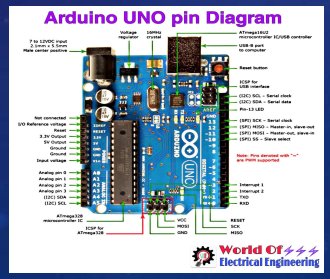
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Methodology:

1.Research and Design:
1. Review existing technologies related to human-following robots and their implementations.
2. Design the robot's layout, selecting appropriate components based on functionality and compatibility.

2.Hardware Implementation:
1. **Microcontroller**: Utilize an Arduino for processing and control tasks.
2. **Sensors**:
   1. **IR Sensors (2)**: For detecting obstacles to the left and right of the robot.
   2. **Ultrasonic Sensor (1)**: For measuring the distance to the target and detecting obstacles in front.
3. **WD Car Kit**: Use the kit's chassis and motors for mobility.
4. **Motor Driver**: Connect the motors to the Arduino through a motor driver to control movement.





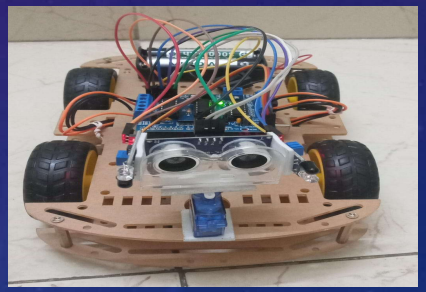
3.Software Development:
1. **Programming Environment**: Use the Arduino IDE for coding the robot's logic.
2. **Detection and Following Logic**:
   1. Implement the logic to follow a target based on distance measured by the ultrasonic sensor.
   2. Use the IR sensors to detect obstacles on the sides and adjust the robot's path accordingly.
3. **Obstacle Avoidance**: Create control algorithms to steer the robot away from detected obstacles while continuing to follow the target.

4.Testing and Evaluation:
1. Conduct tests in various environments to assess performance in tracking and obstacle avoidance.
2. Collect data on the robot's responsiveness, accuracy in following, and ability to navigate around obstacles.
3. Gather user feedback to evaluate the robot's usability and effectiveness.

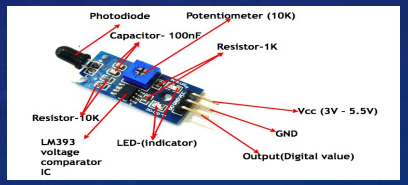
Results

Expected results include:

•**Prototype Functionality**: A working prototype that can autonomously follow a target .
•**Performance Metrics**: Data on tracking accuracy, response time, and obstacle avoidance effectiveness.
•**User Feedback**: Insights from users that can inform future improvements and refinements.







Conclusion

The human-following robot project showcases the potential of robotics in improving daily human activities. By effectively integrating IR and ultrasonic sensors with an Arduino-controlled WD car kit, this robot demonstrates a practical solution for enhancing user interaction and assistance. The findings of this project highlight its effectiveness in real-world applications, and future work could explore features like multi-target tracking or enhanced user controls. This project not only contributes to the field of robotics but also emphasizes the importance of creating user-friendly solutions for everyday challenges.

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